

LECTURE 5

Buoyancy, Flotation and Stability

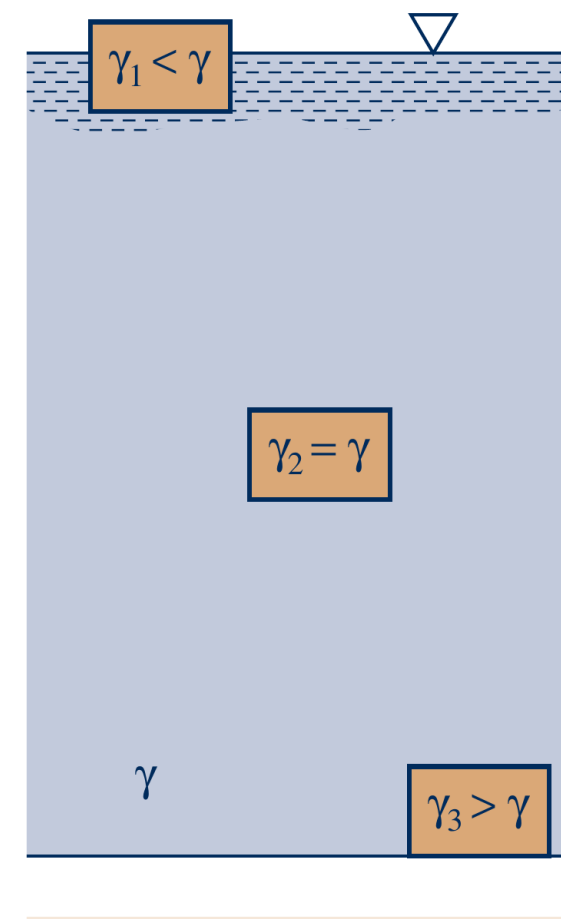
Reference: Munson, Rothmayer, Okiishi & Huebsch, Fundamentals of Fluid Mechanics, Chapter 2
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Buoyancy, flotation and stability

When a body is completely submerged, or floating so that it is only partially submerged, the resultant fluid force on it is called the buoyant force.

The force is upward: pressure increases with depth, depth, so the pressure forces acting on the underside of the body are larger than those acting on its top.

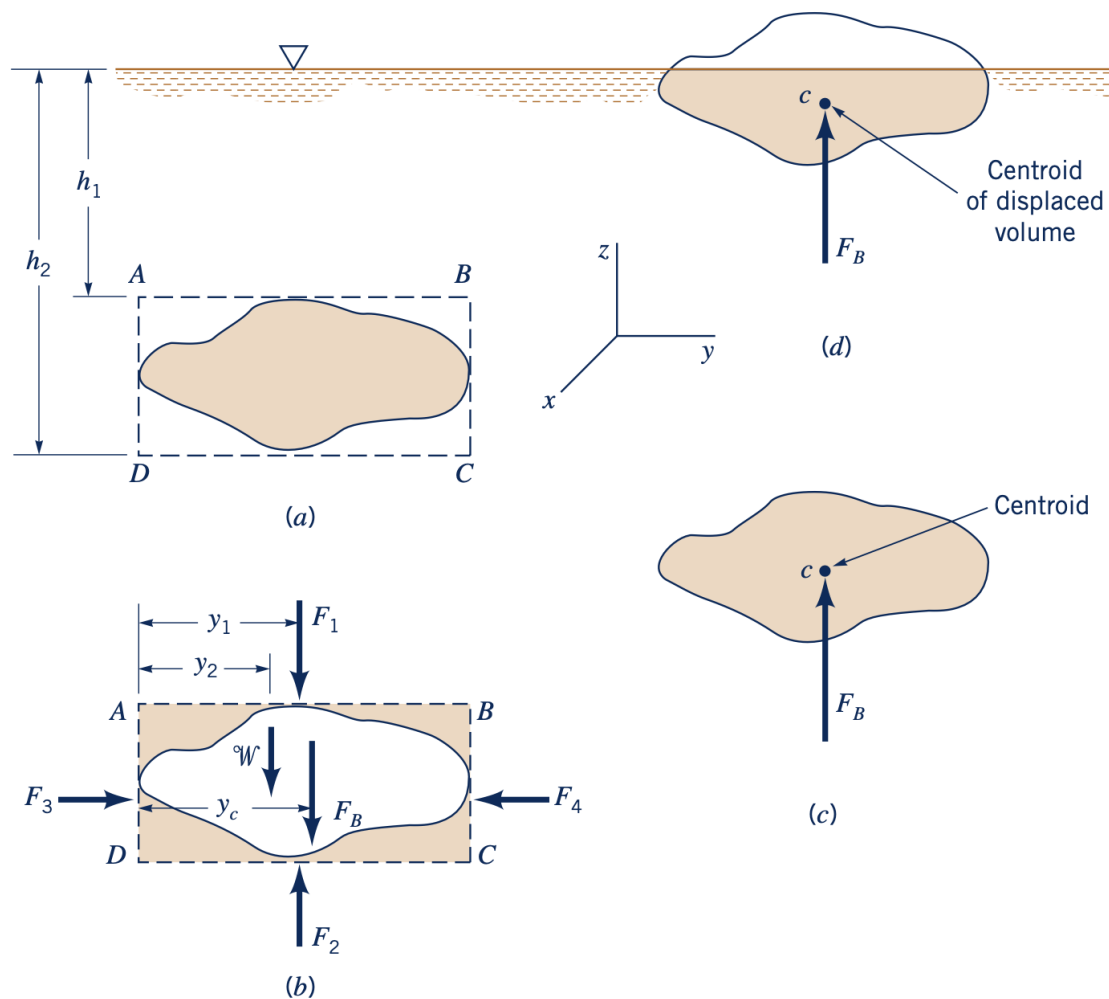
- **Body lighter than the fluid:** it rises and floats
- **Equal specific weights:** it is neutrally buoyant and stays put
- **Body heavier than the fluid:** it sinks



➡ The buoyant force comes from the hydrostatic pressure gradient

The buoyant force: free-body diagram

Buoyant force on a submerged body.



■ **Figure 2.24** Buoyant force on submerged and floating bodies.

Take a body of arbitrary shape and volume $\mathcal{V}$ immersed in a fluid.

Enclose it in a parallelepiped and draw the free-body diagram of the parallelepiped with the body removed.

F_3 and F_4 act on the vertical faces: they are equal and cancel.

$\mathcal{W}$ is the weight of the shaded fluid, the parallelepiped minus the body, and

F_B is the force the body exerts on the fluid.

Equilibrium in z is therefore

$$F_B = F_2 - F_1 - \mathcal{W}$$

Archimedes' principle

If the specific weight of the fluid is constant, the two horizontal faces of the parallelepiped, of area A , carry:

$$F_2 - F_1 = \gamma (h_2 - h_1) A$$

and the weight of the shaded fluid is γ times its volume, the parallelepiped minus the body:

$$F_B = \gamma (h_2 - h_1) A - \gamma [(h_2 - h_1) A - \mathcal{V}]$$

Everything cancels except the body's own volume:

$$F_B = \gamma \mathcal{V}$$

 **The buoyant force equals the weight of the fluid displaced.**

Archimedes' principle states that the buoyant force has a magnitude equal to the weight of the fluid displaced by the body and is directed vertically upward.

Buoyancy: Cartesian Diver



Archimedes' principle states that the **net hydrostatic pressure force** on a submerged or floating body is **equal to the weight of the fluid displaced**.

By pressing the sides of the bottle, the pressure within it is increased and the air within the inverted test tube is compressed.

This causes **the average specific weight of the object (the test tube, air, and the enclosed water) to be greater than that of the surrounding water**.

The tube sinks.

Compressibility and buoyancy: squeezing the bottle compresses the air in the diver, reduces its displaced volume and sinks it.

Where does the buoyant force act?

Summing moments about an axis through D , perpendicular to the diagram:

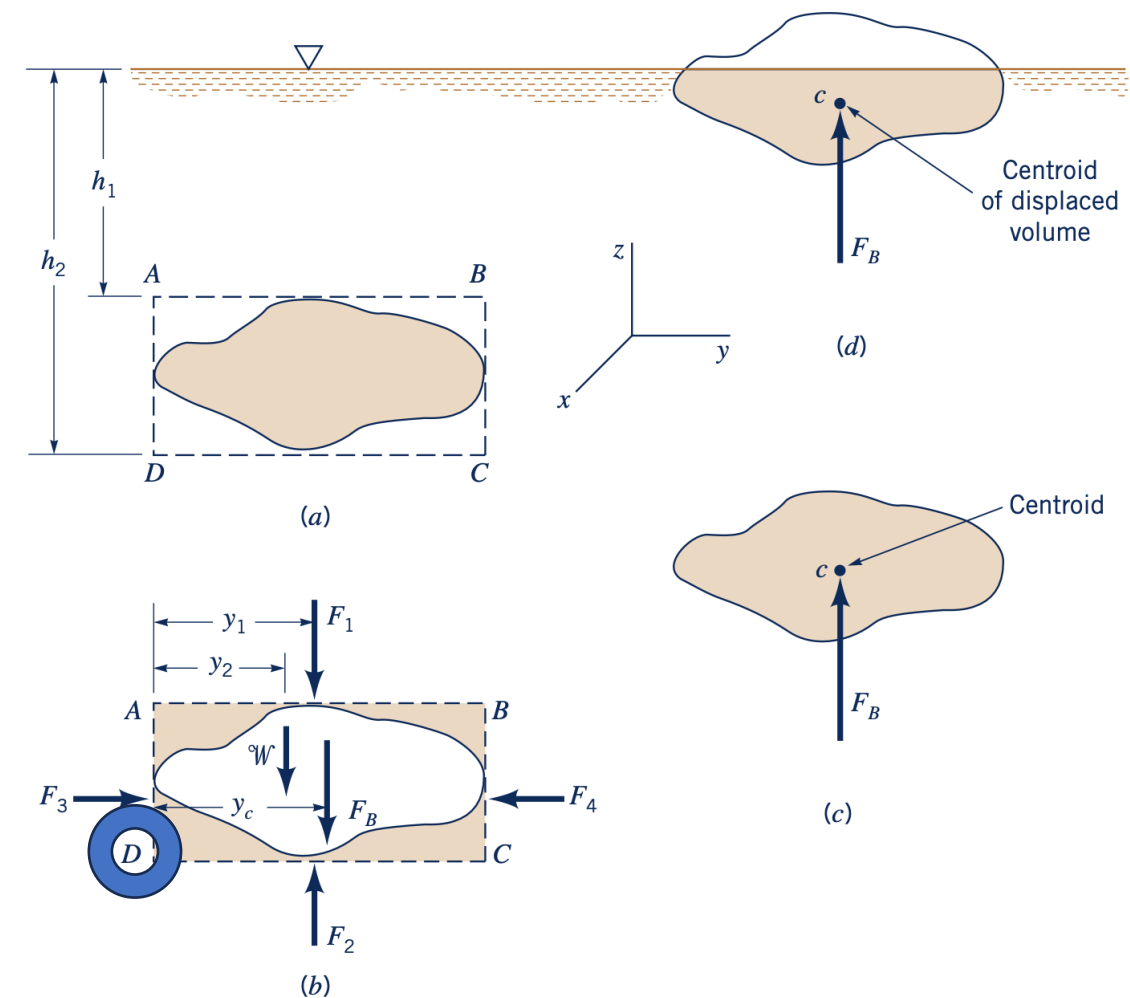
$$F_B y_c = F_2 y_1 - F_1 y_1 - \mathcal{W} y_2$$

and substituting for the forces, with $\mathcal{V}_T$ the total volume of the parallelepiped:

$$\mathcal{V} y_c = \mathcal{V}_T y_1 - (\mathcal{V}_T - \mathcal{V}) y_2$$

The right-hand side is the first moment of the displaced volume about the x - z plane, so y_c is the y coordinate of its centroid. The same holds in x .

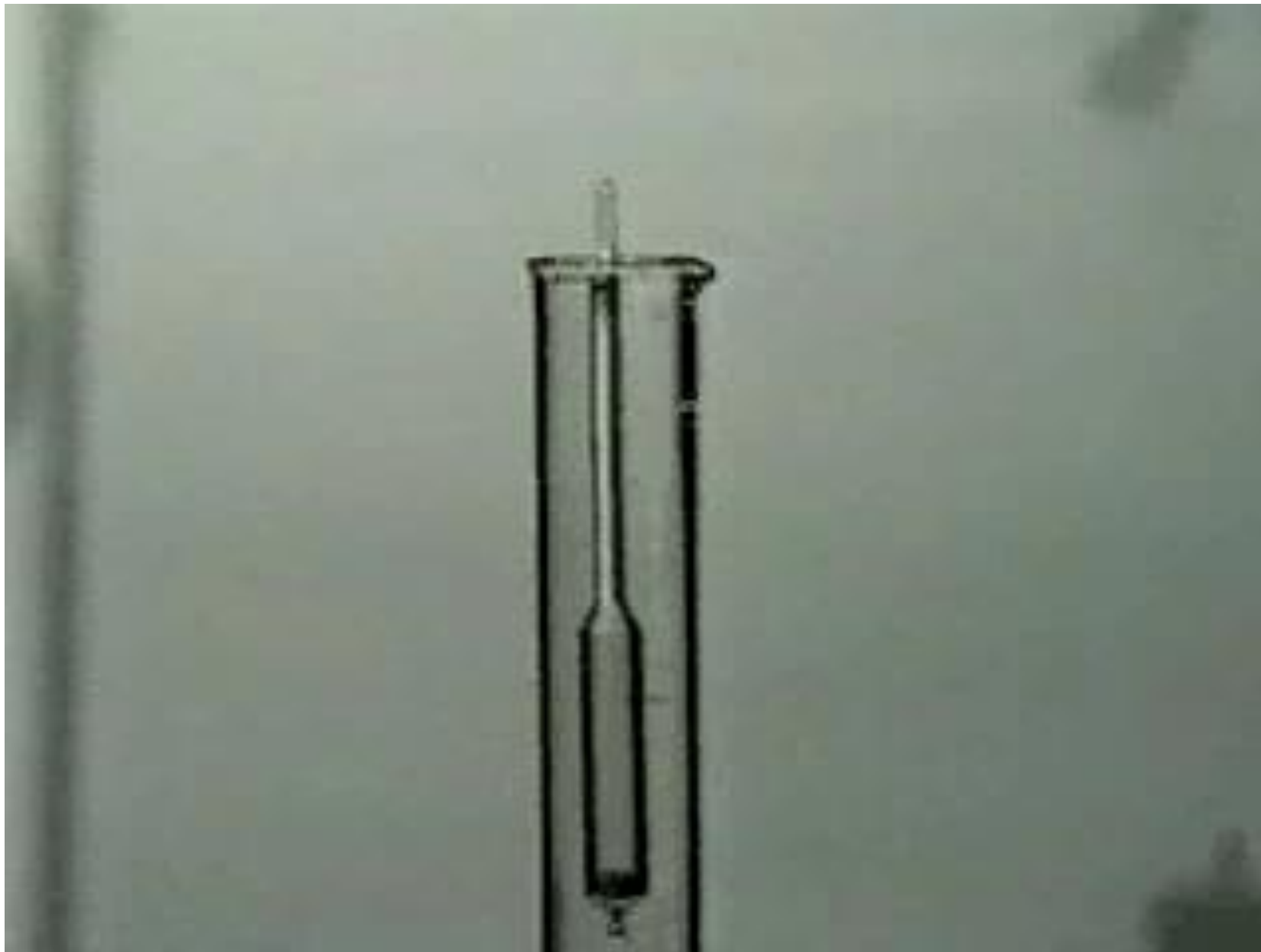
➡ The line of action passes through the centre of buoyancy: *the centroid of the displaced volume.*



■ **Figure 2.24** Buoyant force on submerged and floating bodies.

The buoyant force passes through the centroid of the displaced volume.

Hydrometer



The **specific gravity (SG)** of a liquid can be determined using Archimedes' principle.

A hydrometer is a glass bulb weighted on one end with a narrow stem at the other end.

When placed in a liquid (water) it will float when the constant weight of the hydrometer is balanced by the buoyant force.

Readings along the stem can be related to the specific gravity of the liquid.

If the water is replaced with a heavier liquid (green salt water) the hydrometer will rise since a smaller displaced volume is required to balance the weight.

Floating bodies

The same results hold for a body that is only partially submerged, provided the specific weight of the fluid above the surface is small compared with the liquid.

Since that fluid is normally air, the condition is satisfied in practice.

For a floating body in equilibrium the buoyant force balances the weight, so the displaced volume is fixed by the density ratio:

$$\gamma \mathcal{V}_{\text{disp}} = \mathcal{W}_{\text{body}} \quad \Longrightarrow \quad \frac{\mathcal{V}_{\text{disp}}}{\mathcal{V}_{\text{body}}} = \frac{\rho_{\text{body}}}{\rho_{\text{fluid}}}$$

Calculation for icebergs:

Ice at 917 kg/m³ in seawater at 1025 kg/m³:

$$\frac{\mathcal{V}_{\text{disp}}}{\mathcal{V}_{\text{body}}} = \frac{917}{1025} = 0.89$$

89% of an iceberg is below the surface, and only 11% of it it is what you see.

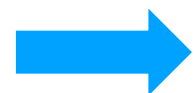
When the specific weight varies with depth

The derivation assumed a constant γ . If the body is immersed in a fluid whose specific weight varies with depth, as in a layered fluid or a stratified ocean:

- **The magnitude of the buoyant force is unchanged:** it is still the weight of the fluid displaced.
- **The line of action moves:** it no longer passes through the centroid of the displaced volume, but through the centre of gravity of the displaced fluid

The distinction matters whenever the stratification is strong:

- in the ocean
- in the atmosphere, and
- in the numerical simulation of any buoyancy-driven flow, where a scheme that does not respect it will drift.



Magnitude from the displaced weight

Line of action from its centre of gravity

Example — buoyant force on a life jacket

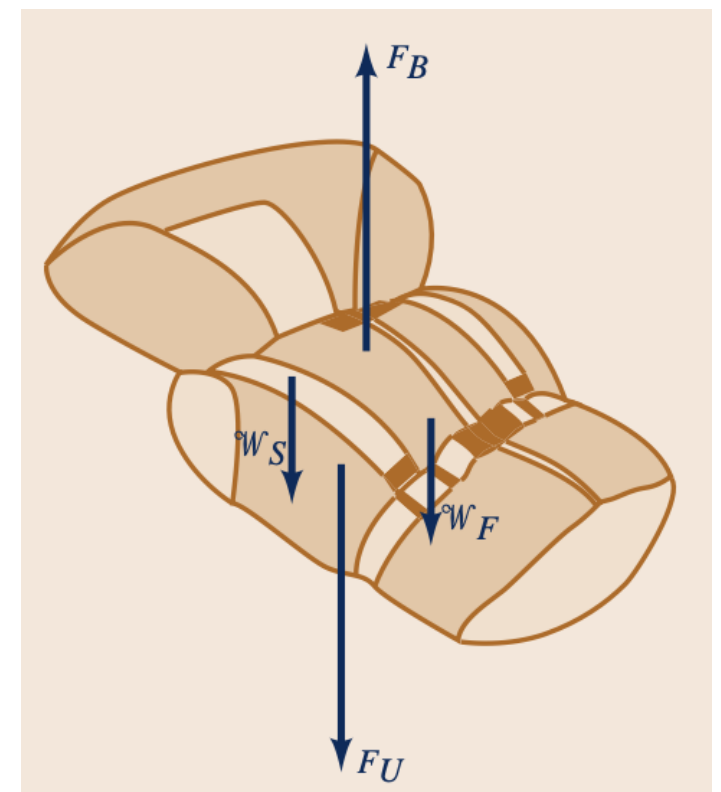
A Type I offshore life jacket must provide a net upward force of at least 98 N on the user. The main flotation is a foam of specific weight 314 N/m³; the remaining material weighs 5.8 N and has negligible volume. Seawater has $\gamma = 10.05 \text{ kN/m}^3$.

Find the minimum volume of foam.

Equilibrium of the life jacket, with the buoyant force:

$$F_B = \mathcal{W}_F + \mathcal{W}_S + F_U, \quad F_B = \gamma_{\text{sw}} \mathcal{V}, \quad \mathcal{W}_F = \gamma_{\text{foam}} \mathcal{V}$$

$$\mathcal{V} = \frac{\mathcal{W}_S + F_U}{\gamma_{\text{sw}} - \gamma_{\text{foam}}} = \frac{5.8 + 98}{10.05 \times 10^3 - 314} = 0.0107 \text{ m}^3$$



Free-body diagram of the life jacket.

Buoyancy within a fluid

Buoyancy is not restricted to a solid body in a fluid. We can have a parcel of fluid rather than a solid. If the parcel has density ρ_2 and the surroundings ρ_1 :

$$\mathcal{W} = \rho_2 g \mathcal{V}, \quad F_B = \rho_1 g \mathcal{V} \quad \implies \quad F_{\text{net}} = (\rho_1 - \rho_2) g \mathcal{V}$$

- $\rho_2 = \rho_1$: weight and buoyant force balance exactly, and there is no net force
- $\rho_2 < \rho_1$: the parcel accelerates upward
- $\rho_2 > \rho_1$: it accelerates downward

The density difference may come from two different fluids, or from a temperature difference within one fluid.

Smoke rises from a fire because it is hotter, and therefore lighter, than the surrounding air.

Example: atmospheric clouds



Mushroom-shaped cloud produced by an atomic bomb blast -> large buoyant force on the very hot, light weight air produced by the blast.

The generation of cumulus clouds (those puffy, cotton-ball, summer-time clouds that dot the noon-time sky) are produced similarly, although with considerably less energy.

Clouds are often a result of buoyant effects in the atmosphere. As light weight, warm humid air rises it cools, and the moisture condenses as clouds.

Conceptually, the rise of this hot air is similar to the rise of a ping-pong ball that is released underwater.

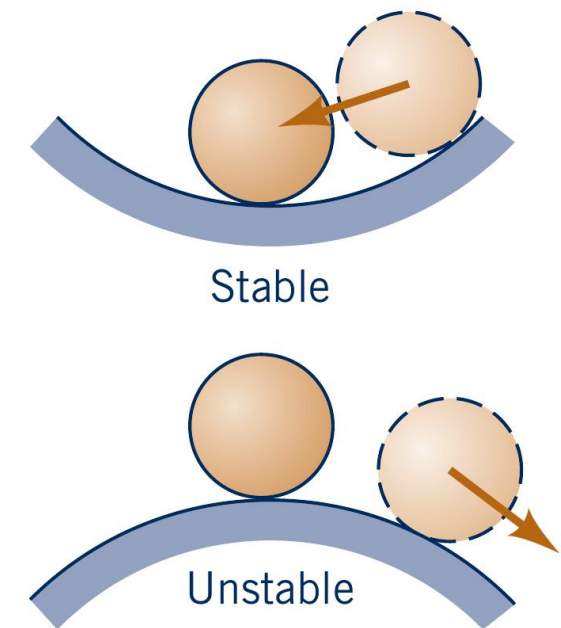
Stability

A body is in a stable equilibrium position if, when displaced, it returns to that position.

It is in an unstable equilibrium position if, when displaced even slightly, it moves to a new one.

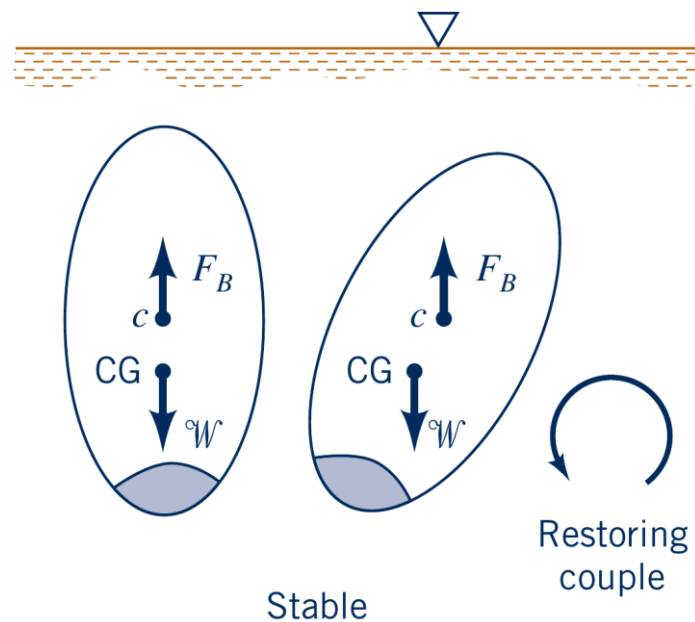
For a submerged or floating body the question is not trivial, because the centre of buoyancy and the centre of gravity do not in general coincide.

A small rotation then produces a couple, and the sign of sign of that couple decides the outcome.

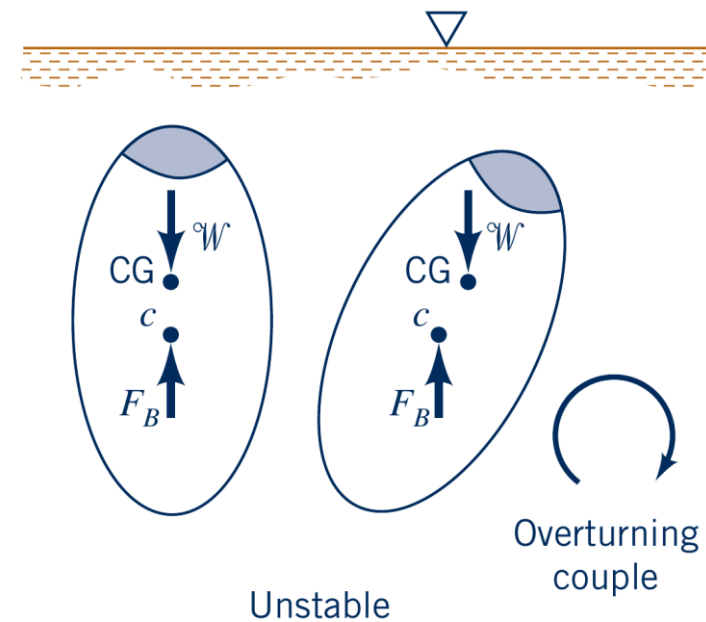


➡ Displace the body and ask which way the couple turns it

Completely submerged bodies



Centre of gravity below the centre of buoyancy: stable.

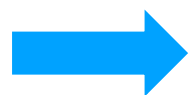


Centre of gravity above it: unstable.

For a fully submerged body the displaced volume never changes, so the centre of buoyancy is fixed in the body.

Rotating the body then leaves $\mathcal{W}$ and F_B on parallel lines a small distance apart, and the couple restores or overturns according to which point is on top.

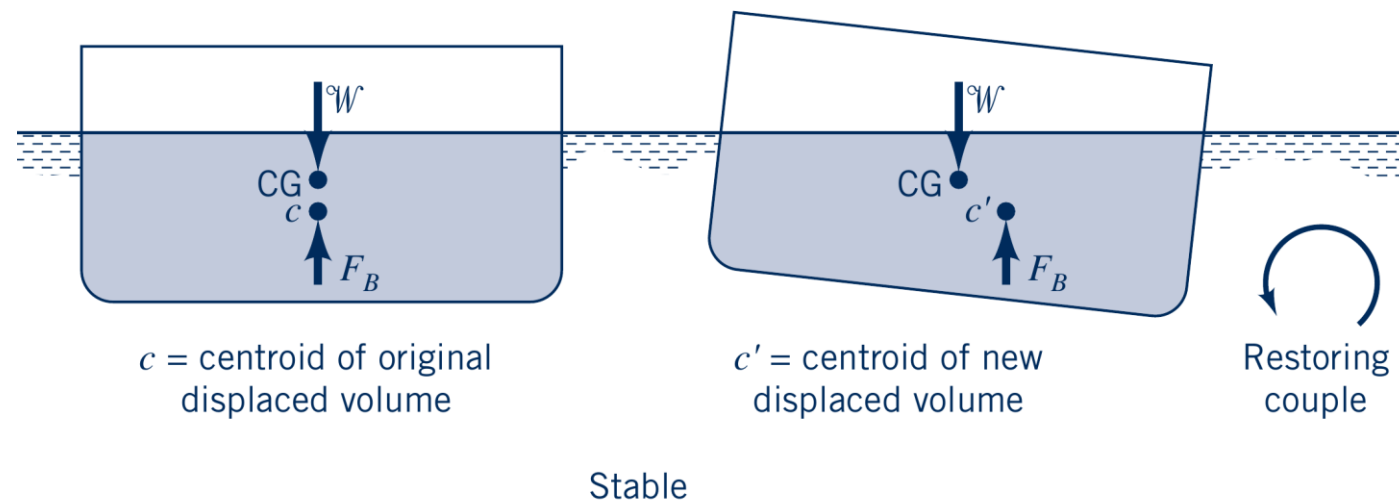
submerged: CG below $B \Rightarrow$ stable



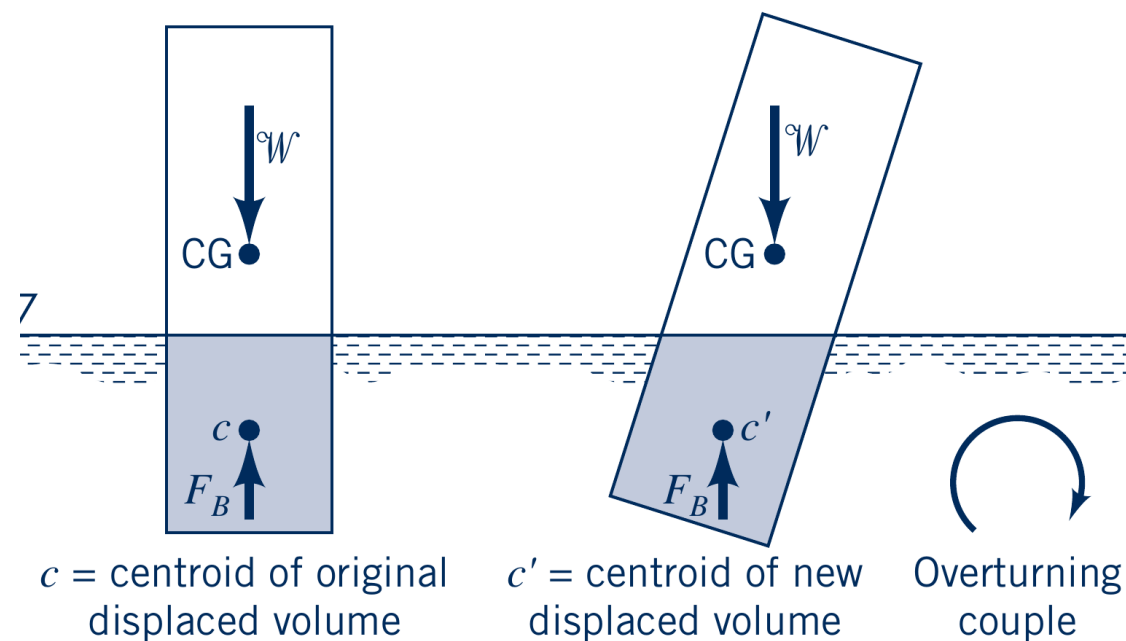
Submarines, balloons and bathyscaphes are equipped with heavy weights or ballast tanks to control up-and-down movement for exactly this

Floating bodies: the centre of buoyancy moves

For a floating body the problem is harder, because as the body heels the shape of the displaced volume changes and the centre of buoyancy shifts.



*A barge that rides low in the water is stable even with its centre of gravity above the centre of buoyancy: the shifted F_B and W form a **restoring couple**.*



For a tall, slender body the same small rotation gives an overturning couple.

The metacentre

The standard way to make it quantitative is the metacentre M :

the point where the line of action of F_B , after a small heel, crosses the body's own centreline.

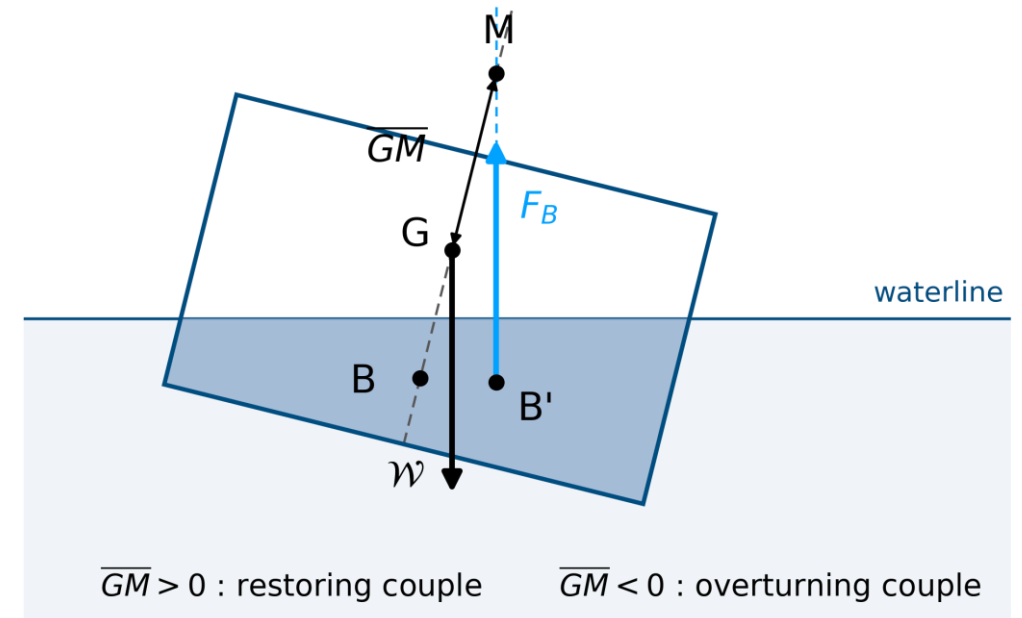
$$\overline{GM} = \frac{I_{yy}}{V_{\text{disp}}} - \overline{BG}$$

The numerator is the second moment of area of the waterplane about the axis of heel, and BG is the distance from B to G , positive when G is above B .

$$\overline{GM} > 0 \Rightarrow \text{stable}, \quad \overline{GM} < 0 \Rightarrow \text{unstable}$$

$$I_{yy} = \frac{L b^3}{12}$$

For a rectangular waterplane of length L and beam b . The beam enters cubed: a wide, shallow hull is far stiffer than a narrow one.



The body heeled through a small angle.

Geometry is important

Both of these examples are stable, but not equally so.
The kayak is narrow:

- **The kayak is narrow:** its waterplane moment is small, GM is barely positive, and very little is needed to roll it.

- **The houseboat is wide and shallow:** GM is comfortable.

Determining stability in general is difficult, because the answer depends in a complicated way on the geometry and on the distribution of weight.

Wind gusts, currents and free surfaces inside the hull make it harder still.

It is a central problem in the design of ships, submarines and bathyscaphes, and the working subject of naval architecture.



Marginally stable

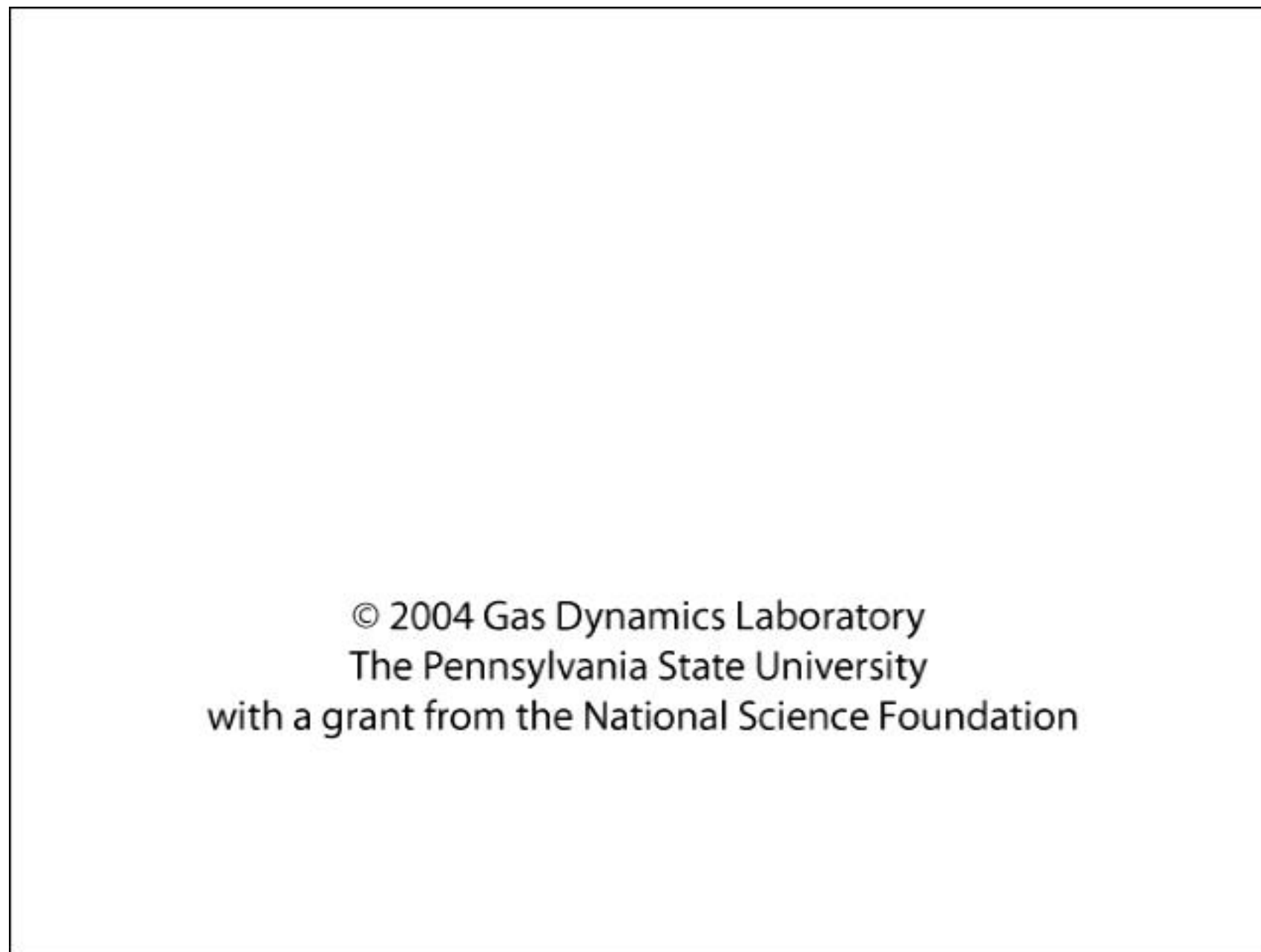


Very stable

Density differences in fluids

The effects of buoyancy can also be observed in fluid-fluid interactions. The only requirement is that density differences exist between the fluids.

The video shows several examples of a fluid rising or falling due to the density differences.



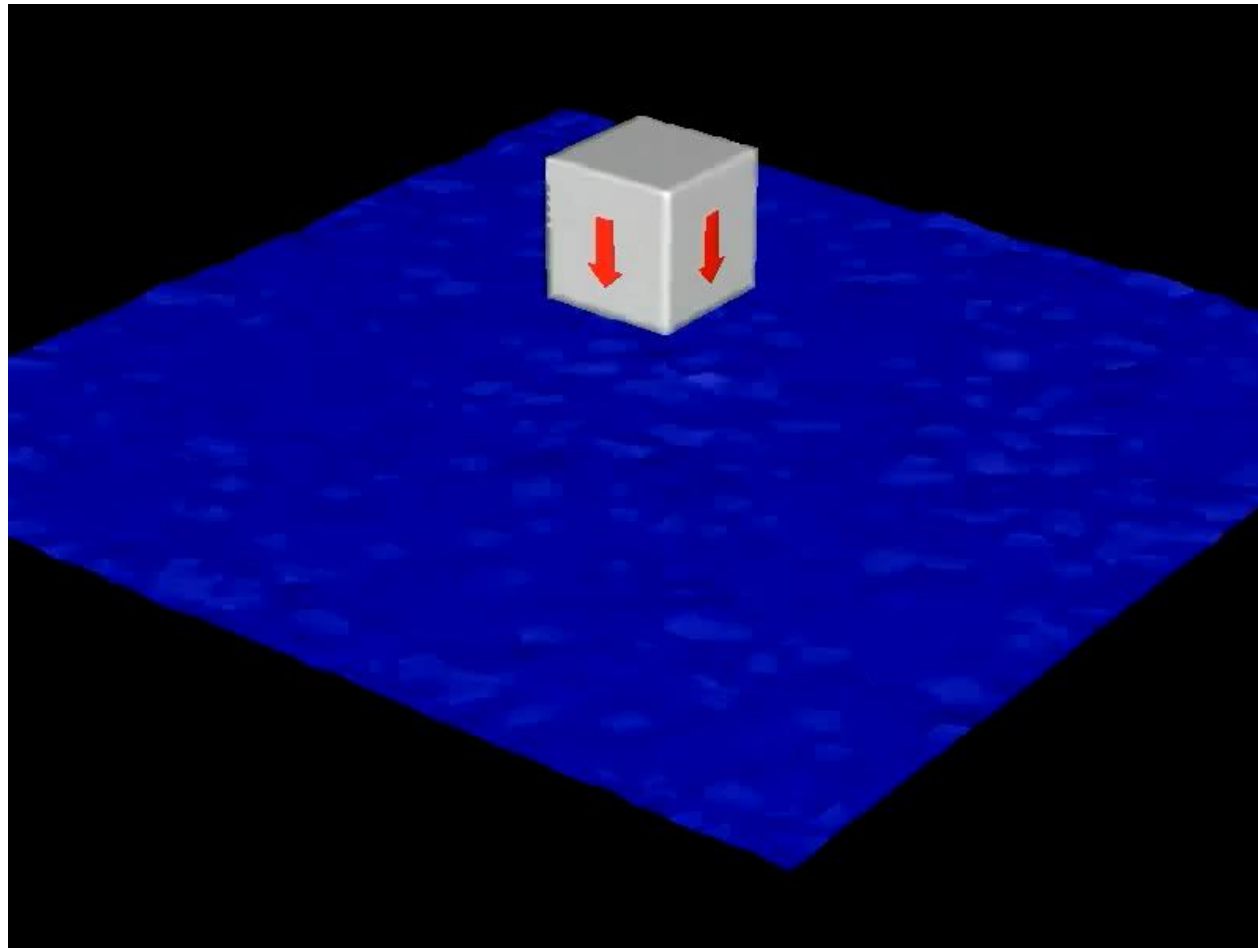
1. Turbulent plume rising from the flame of a cigarette lighter

2. Cigarette lighter, but with no flame: butane is released.

3. Liquid acetone is placed in a beaker which produces a dense vapor.

4. CFD results simulate an open freezer with cold air falling to the floor, since the cold air is more dense than the room air.

Stability of a floating cube



An object is said to be in a stable position if, when displaced, it returns to that position.

The animation shows a 3D box being dropped into a tank of water and eventually obtaining its stable orientation.

This is a physics-based animation derived from a CFD simulation of the given flow geometry.

The flow field was broken into 380,000 small small cells and the equations governing the the fluid motion were solved numerically.

Because the flow is unsteady, the time interval shown was broken into 18,000 small small time steps, with calculations done at each one. Considerable computational power power is needed for such calculations.

The object rolls to sit on a face or on an edge depending on the density ratio.

Stability of a model barge



In this model the centre of gravity can be raised or lowered by moving the weights on the mast, while the centre of buoyancy stays below the free surface

With a low centre of gravity the object is very stable. As the centre of gravity is raised it becomes more tippy, and eventually it becomes unstable.

A floating body can be stable even though its centre of gravity is located above its centre of buoyancy.

Lecture summary

Buoyancy is the hydrostatic equation integrated over a closed surface. It collapses to a single force, equal to the weight of the fluid displaced and acting through the centroid of the displaced volume. Stability asks where that force goes once the body is tilted.

$$F_B = \gamma \mathcal{V}$$

Archimedes' principle

$$F_B = W$$

floating equilibrium

$$\frac{\mathcal{V}_{\text{sub}}}{\mathcal{V}} = \frac{\rho_{\text{body}}}{\rho_{\text{fluid}}}$$

floating: the density ratio

$$\mathbf{r}_B = \frac{1}{\mathcal{V}} \int_{\mathcal{V}} \mathbf{r} d\mathcal{V}$$

the centre of buoyancy

$$F_B = \int_{\mathcal{V}} \gamma(z) d\mathcal{V}$$

when the fluid is stratified

$$z_{CG} < z_{CB}$$

submerged: CG below CB

$$\overline{GM} = \frac{I_{yy}}{\mathcal{V}_{\text{sub}}} - \overline{BG}$$

the metacentric height

$$\overline{GM} > 0$$

stable floating